

Homework (Carolina Reaper)

- Q1.** A music producer is creating a rhythm pattern with 12 beats. If the pattern is divided into groups of 4 beats, what is the remainder?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q2.** A painting has a canvas size of 45 inches by 30 inches. If the painter wants to divide the canvas into equal sections with a width of 5 inches, what is the remainder of the division of the length?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q3.** A frequency of 36 Hz is divided by 4 Hz. What is the quotient and remainder?
(A) 9 R 0
(B) 8 R 2
(C) 7 R 3
(D) 6 R 4
(E) 5 R 5
- Q4.** The number 48 is divisible by 2, 3, and 5. What is the smallest number that is divisible by 2, 3, 5, and 9?
(A) 90
(B) 180
(C) 270
(D) 360
(E) 450
- Q5.** A song has 63 beats per minute. If the song is divided into sections of 7 beats, what is the remainder?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q6.** The sum of the digits of 216 is 9. What is the largest number less than 1000 that has a digit sum of 9?
(A) 900
(B) 810
(C) 729
(D) 630
(E) 540
- Q7.** A painter is mixing colors with frequencies of 24 Hz, 30 Hz, and 36 Hz. What is the greatest common divisor of these frequencies?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q8.** A music composer is creating a rhythm pattern with 50 beats. If the pattern is divided into groups of 8 beats, what is the remainder?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 6

Open Ended Questions (FRQ)

- Q9.** A painter is mixing colors with frequencies of 80 Hz, 100 Hz, and 120 Hz. Perform long division to find the quotient and remainder when 240 is divided by the least common multiple of these frequencies. Show your work.
- Q10.** The length of a painting is 135 inches. If the width is 90 inches, what is the remainder when the area of the painting is divided by 5? Express your answer as $a \cdot b + r$, where a is the quotient, b is the divisor, and r is the remainder. Show your work.

Chef's Tips

- Tip 1: Make sure to check the divisibility rules for 2, 3, 5, 9, and 10.
- Tip 2: Use the division algorithm to find the quotient and remainder.
- Tip 3: Use the least common multiple (LCM) to find the smallest number that is divisible by multiple numbers.